Containerised Pipelines for Sensitive Sequencing Data

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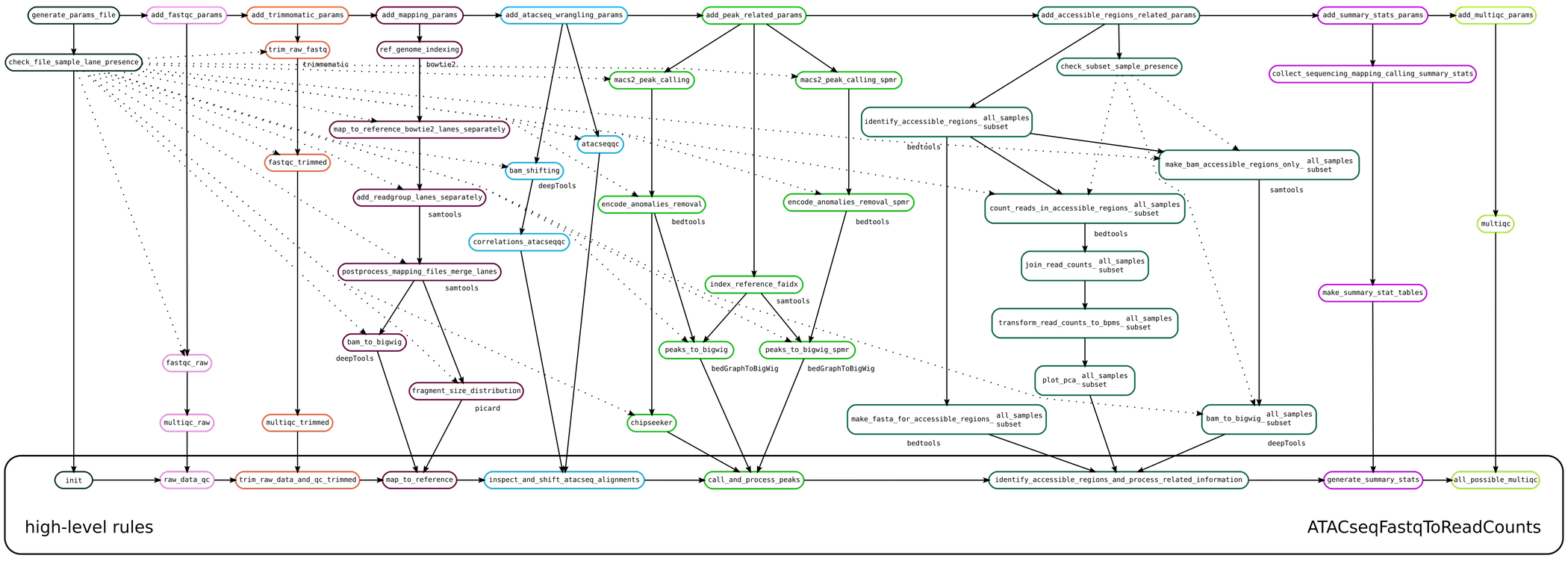
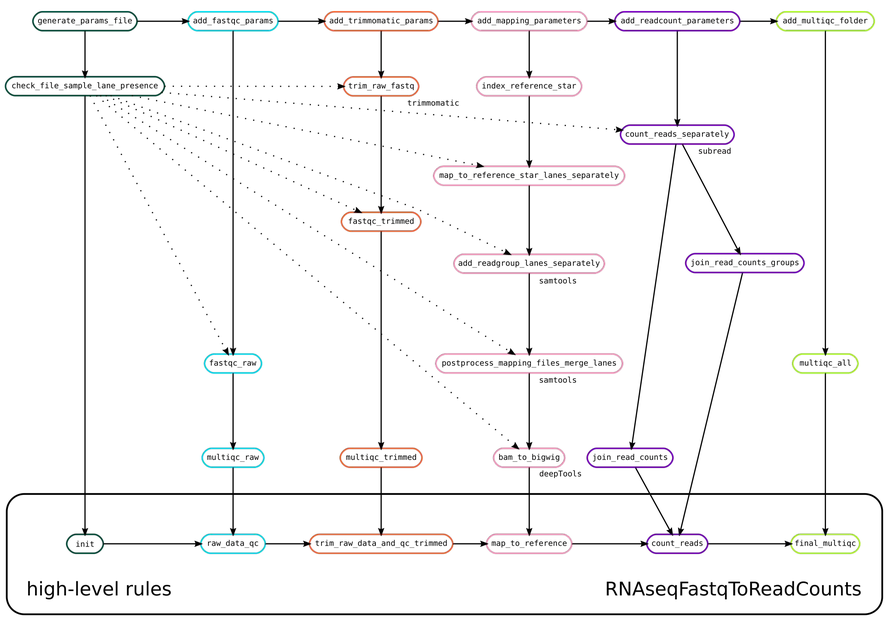
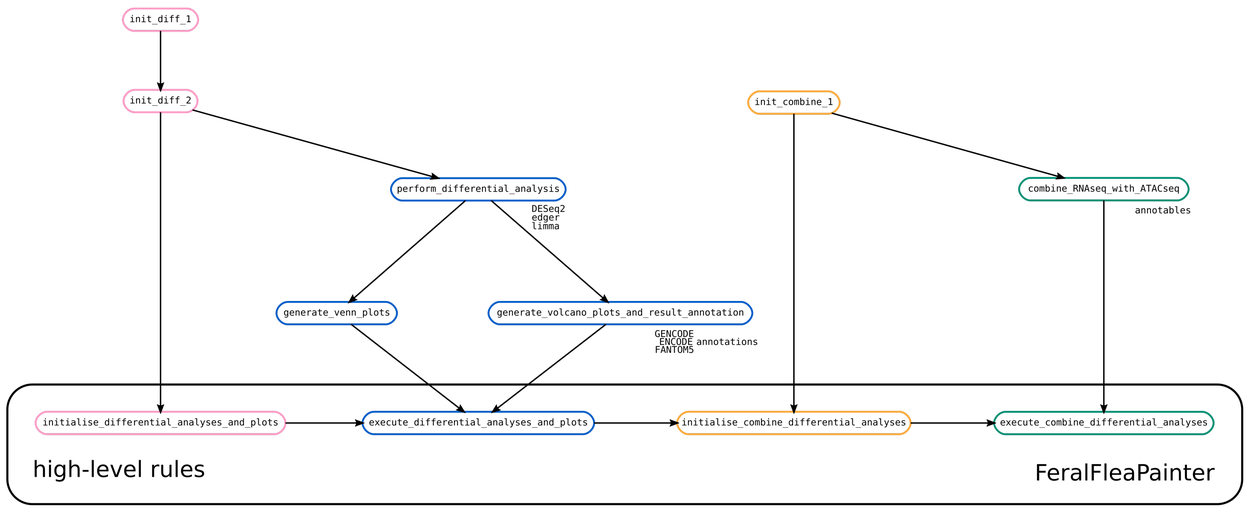
Snakemake Pipelines







Features

- All the analyses are performed on an HPC dedicated to work with sensitive data based in Norway and called TSD. The pipelines aim on being FAIR.
- The pipelines are orchestrated with `Snakemake`[1].
- `Singularity`[2] container images with installed bioinformatic tools are used to execute the conventional bioinformatic software instead of installing it on the underlying operating system. All the pipeline scripts are kept outside of the container images for more comfortable code development and seamless future adjustments.
- Paired nature of the generated samples is utilised in the third pipeline, the first two pipelines (processing raw sequencing reads) are independent of the experiment design used for the data at hand.
- All the user adjustable parameters are provided to the pipelines via config files.
- The user executes merely a sequence of the high-level rules.
- Automated lane merging in the first two pipelines.

Limitations

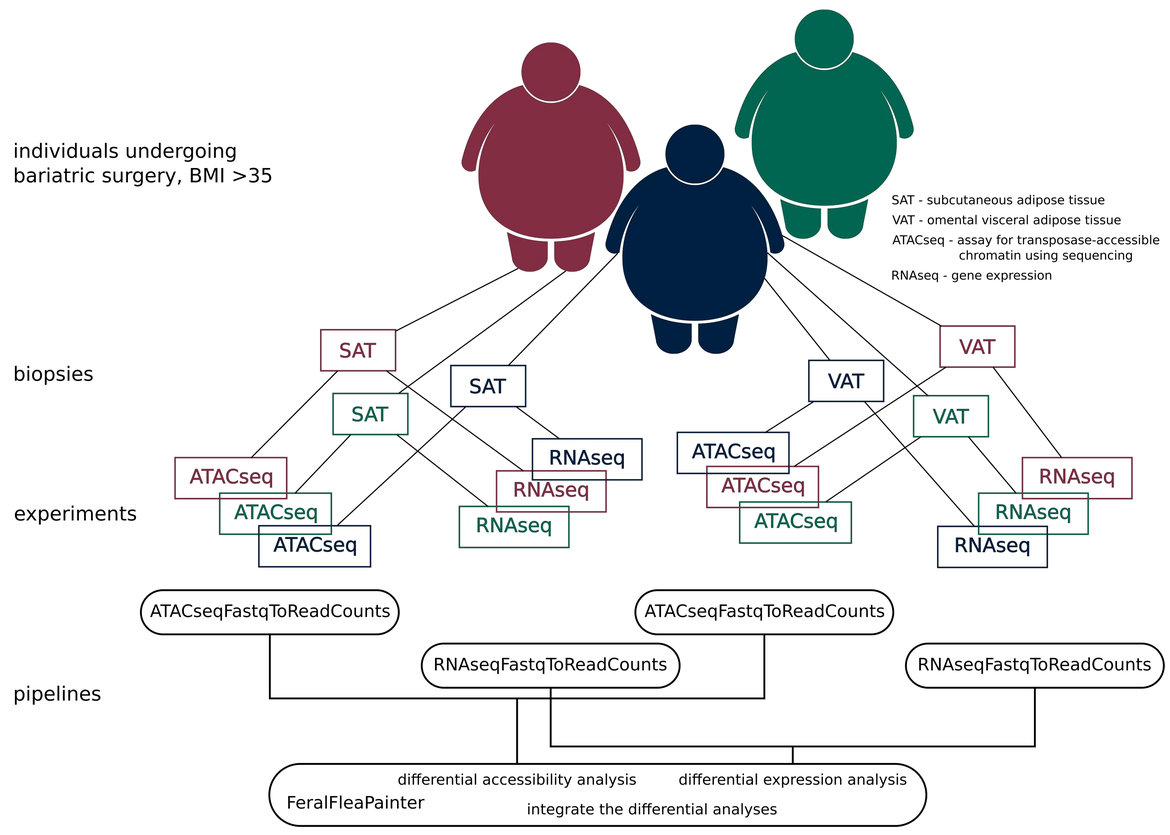
- Container images are too big to be naïvely shared via LFS.
- Paired-end Illumina sequencing reads only.

Code Availability

`https://gitlab.com/EpiGen/<pl>`, where

`<pl>` in (ATACseqFastqToReadCounts, RNAseqFastqToReadCounts, FeralFleaPainter)

Samples & Generated Data



Acknowledgments

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Used Tools

[1] https://snakemake.readthedocs.io
[2] https://sylabs.io/

fastqc, multiqc, trimmomatic, bowtie2, samtools, deepTools, picard, atacseqqc, macs2, bedtools, bedToGraphBigwig, star, subread, DESeq2, edger, limma, gencode, encode, fantom5, annotables

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